

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0620

Fairfield 138 kV
July 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0620. The Interconnection Customer proposes a 192.6 MW solar photovoltaic facility interconnecting at the Fairfield 138 kV substation in the Virginia area. The request was received on May 22, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 193 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,658,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dpc Prairie Du Chien / Frito Lay 138 kV line
Contingency: Purdy / Chatom 138 kV line
Loading: 133.0% (Rating: 233 MVA)
DFAX: 6.8%
Pre-project loading: 127.4%
2. Facility: Aspen / Villa Mar 2 1658 138 kV line
Contingency: Crandall / Aspen 345 kV
Loading: 131.1% (Rating: 227 MVA)
DFAX: 22.0%
Pre-project loading: 112.4%
3. Facility: Carodean / Crandall 138 kV line
Contingency: Crandall / Dayton 345 kV
Loading: 130.7% (Rating: 236 MVA)
DFAX: 8.9%
Pre-project loading: 123.3%
4. Facility: Harrisonbug / Sans Souci 138 kV
Contingency: Dayton / Chatom 345 kV
Loading: 127.2% (Rating: 254 MVA)
DFAX: 6.1%
Pre-project loading: 122.5%
5. Facility: Oak Crk Sta / Sans Souci 138 kV
Contingency: Chatom / Dayton Substation 345 kV Line
Loading: 126.8% (Rating: 255 MVA)
DFAX: 6.1%
Pre-project loading: 122.1%

6. Facility: Redland / Carodean 138 kV
Contingency: Crandall / Dayton 345 kV
Loading: 126.5% (Rating: 222 MVA)
DFAX: 8.9%
Pre-project loading: 118.7%
7. Facility: Oak Crk Sta / Wadsworth 138 kV
Contingency: Chatom / Dayton 345 kV
Loading: 123.8% (Rating: 230 MVA)
DFAX: 6.1%
Pre-project loading: 118.7%
8. Facility: Weyers Cave / Grottoes 138 kV
Contingency: Crandall / Dayton Substation 345 kV
Loading: 121.3% (Rating: 352 MVA)
DFAX: 8.6%
Pre-project loading: 116.6%
9. Facility: Dayton 345/138kV TX
Contingency: Dayton Substation / Chatom 345 kV
Loading: 116.9% (Rating: 479 MVA)
DFAX: 12.3%
Pre-project loading: 112.0%
10. Facility: Weyers Cave / Greenbank 138 kV line
Contingency: Crandall / Dayton 345 kV Line
Loading: 116.5% (Rating: 265 MVA)
DFAX: 8.6%
Pre-project loading: 110.3%
11. Facility: Villa Mar 2 1658 / Orchard 138 kV L
Contingency: Crandall / Aspen 345 kV L
Loading: 113.6% (Rating: 249 MVA)
DFAX: 22.0%
Pre-project loading: 96.6%
12. Facility: Chatom 345 kV/138 kV XFMR
Contingency: Dayton Substation 345 kV/138 kV Transformer
Loading: 112.0% (Rating: 494 MVA)
DFAX: 12.1%
Pre-project loading: 107.3%
13. Facility: Declo / Orchard 138 kV Line
Contingency: Crandall / Aspen 345 kV L
Loading: 110.8% (Rating: 244 MVA)
DFAX: 22.0%
Pre-project loading: 93.4%
14. Facility: Dayton / Harrisonburg 138 kV
Contingency: Dayton / Chatom 345 kV
Loading: 108.9% (Rating: 487 MVA)
DFAX: 12.3%
Pre-project loading: 104.0%
15. Facility: Aspen / Fairfield 138 kV
Contingency: Aspen Substation / Lowmoor 345 kV
Loading: 103.4% (Rating: 214 MVA)
DFAX: 75.2%
Pre-project loading: 35.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dpc Prairie Du Chien / Frito Lay 138 kV	Rebuild line with higher-capacity conductor	\$4,712,000	41% (\$1,930,000)
Aspen Substation / Villa Mar 2 1658 138 kV	Rebuild line with higher-capacity conductor	\$1,588,000	37% (\$590,000)
Carodean / Crandall 138 kV line	Rebuild line with higher-capacity conductor	\$3,166,000	16% (\$501,000)
Harrisonbug / Sans Souci 138 kV	Rebuild line with higher-capacity conductor	\$2,931,000	39% (\$1,143,000)
Oak Crk Sta / Sans Souci 138 kV	Rebuild line with higher-capacity conductor	\$1,846,000	39% (\$720,000)
Redland / Carodean 138 kV	Rebuild line with higher-capacity conductor	\$1,332,000	16% (\$211,000)
Oak Crk Staion / Wadsworth 138 kV line	Rebuild line with higher-capacity conductor	\$7,998,000	39% (\$3,118,000)
Grottoes / Weyers Cave 138 kV	Rebuild line with higher-capacity conductor	\$7,351,000	13% (\$943,000)
Dayton 345/138kV TX	Replace transformer with higher-rated unit	\$16,327,000	27% (\$4,393,000)
Weyers Cave / Greenbank 138 kV Line	Rebuild line with higher-capacity conductor	\$6,616,000	13% (\$863,000)
Villa Mar 2 1658 / Orchard 138 kV L	Rebuild line with higher-capacity conductor	\$5,194,000	37% (\$1,930,000)
Chatom 345-138 kV TX	Replace transformer with higher-rated unit	\$13,632,000	24% (\$3,327,000)
Declo / Orchard 138 kV Line	Rebuild line with higher-capacity conductor	\$8,783,000	37% (\$3,263,000)
Harrisonburg Substation / Dayton 138 kV	Reconductor line	\$613,000	27% (\$165,000)
Aspen / Fairfield 138 kV	Reconductor line	\$5,624,000	81% (\$4,561,000)
	Total allocated to project		\$27,658,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.